



Lab 03: Cloud IAM Qwik Start

Google Cloud Associate Cloud Engineer | GSP064

Overview

Explored Google Cloud Identity and Access Management (IAM) by signing in as two different users with different roles. Experienced how granting and revoking permissions affects what each user can see and do in the GCP Console.

IAM Roles Covered

Role	Permissions
roles/owner	Full access including managing roles, permissions, and billing
roles/editor	All viewer permissions plus ability to modify existing resources
roles/viewer	Read-only access, cannot modify resources or state
roles/browser	Read access to browse resource hierarchy only
Storage Object Viewer	Read access to objects in a specific Cloud Storage bucket only

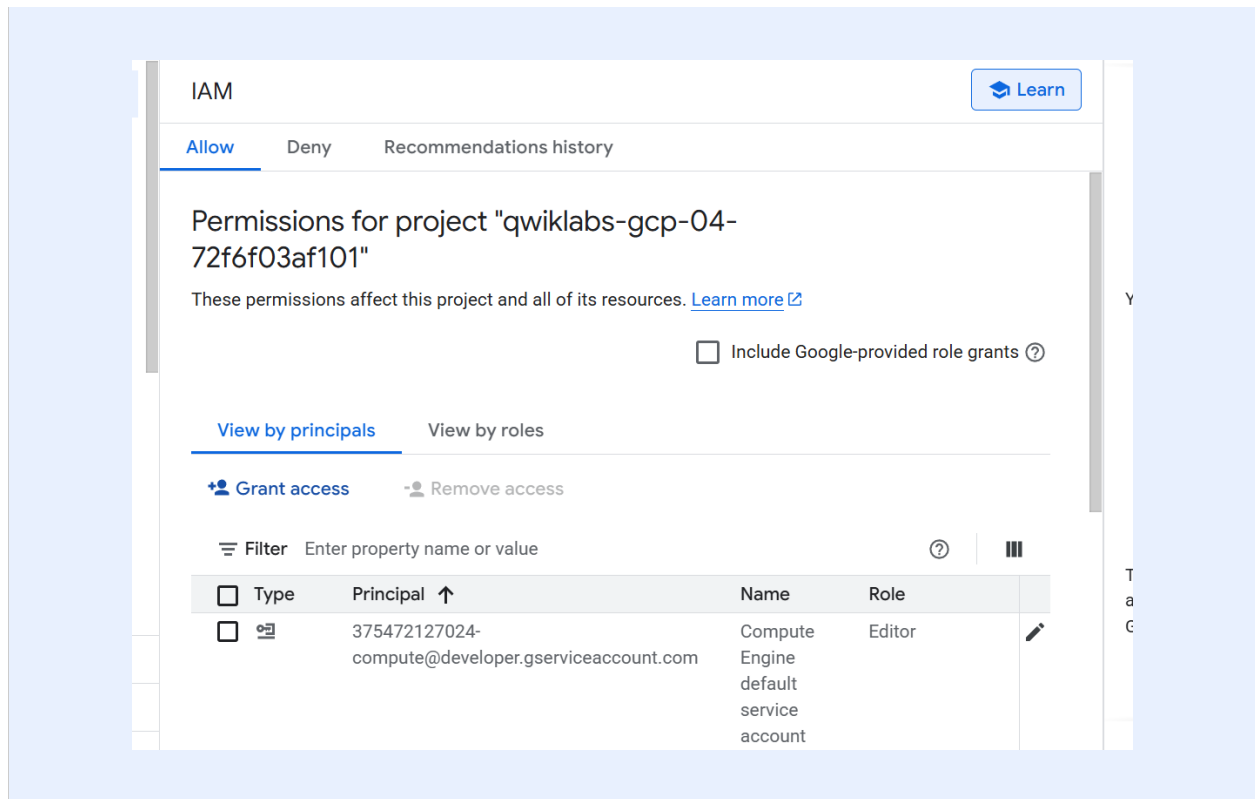
Objectives

- Explore IAM console and project level roles
- Create a Cloud Storage bucket and upload a file as Owner
- Verify Viewer access as Username 2
- Remove project access from Username 2
- Grant Storage Object Viewer role to Username 2
- Verify specific Cloud Storage access via CLI

Task 1 - Explore IAM Console and Roles

Signed in as Username 1 (Owner) and Username 2 (Viewer) in separate incognito windows. Explored the IAM console to compare what each user can see and do.

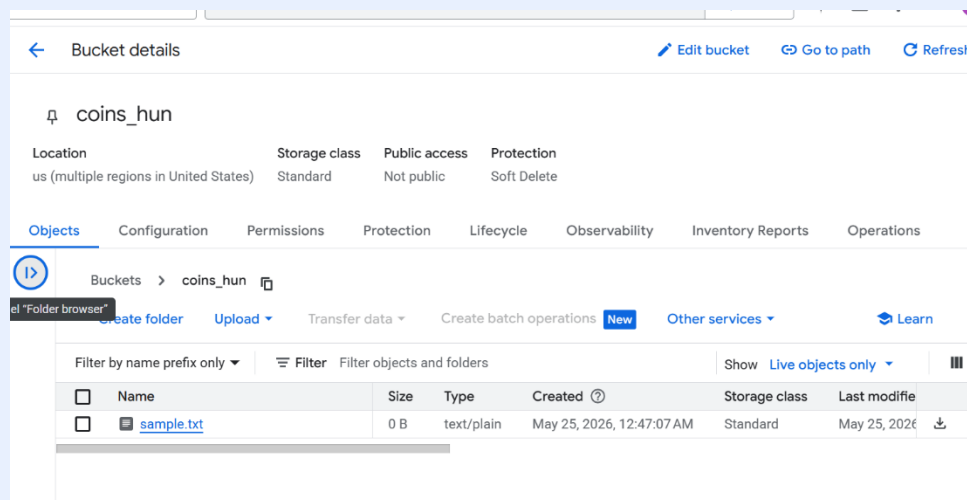
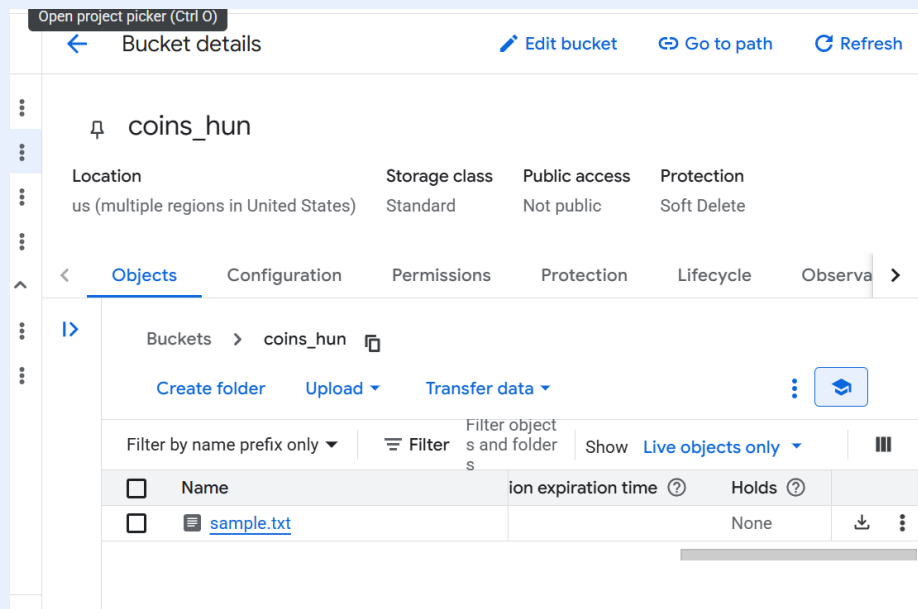
- Username 1 had Owner role — could grant and manage permissions
- Username 2 had Viewer role — Grant Access button was grayed out



Task 2 - Create a Bucket and Upload a File

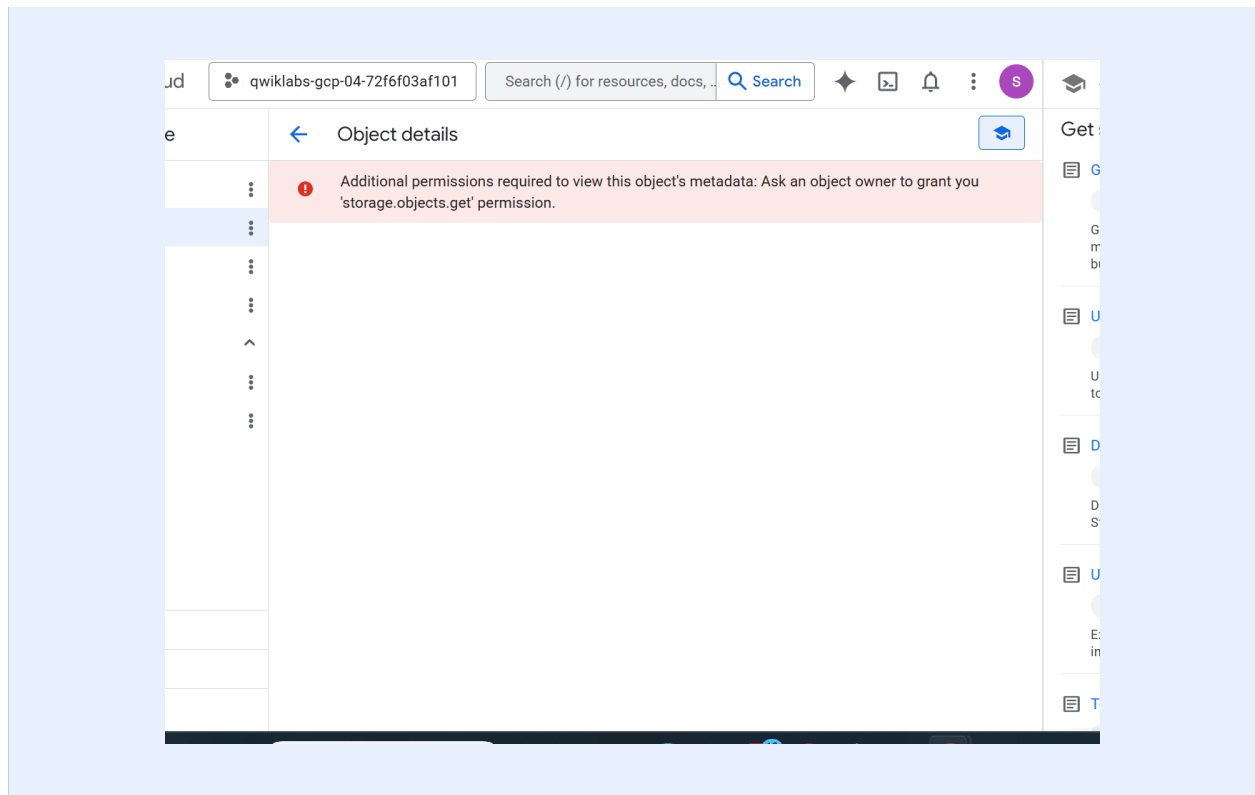
As Username 1 (Owner), created a Cloud Storage bucket with Multi-Region location and uploaded a sample.txt file. Then verified Username 2 could see the bucket as a Viewer.

- Location type: Multi-Region
- Uploaded and renamed file to sample.txt



Task 3 - Remove Project Access

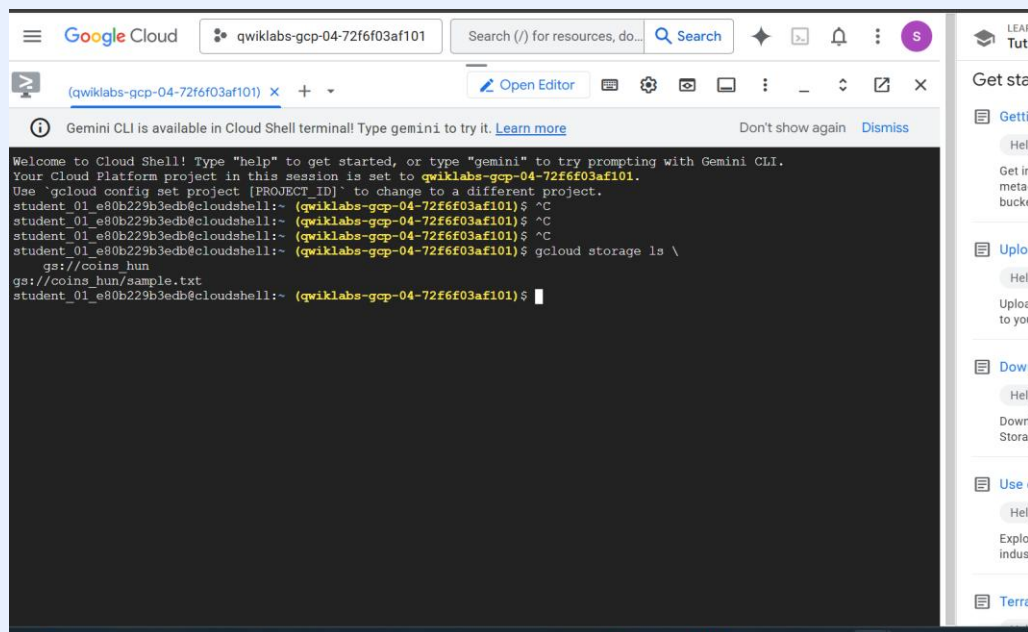
As Username 1, removed the Viewer role from Username 2 via IAM & Admin. After removal Username 2 received a permissions error when trying to access Cloud Storage.



Task 4 - Add Cloud Storage Permissions

Granted Username 2 the Storage Object Viewer role at the bucket level instead of project level. This gives specific access to Cloud Storage only without project-wide access.

```
gcloud storage ls gs://[YOUR_BUCKET_NAME]
```



```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-04-72f6f03af101.
Use 'gcloud config set project [PROJECT ID]' to change to a different project.
student_01_e80b229b3edb@cloudshell:~ (qwiklabs-gcp-04-72f6f03af101) $ ^C
student_01_e80b229b3edb@cloudshell:~ (qwiklabs-gcp-04-72f6f03af101) $ ^C
student_01_e80b229b3edb@cloudshell:~ (qwiklabs-gcp-04-72f6f03af101) $ ^C
student_01_e80b229b3edb@cloudshell:~ (qwiklabs-gcp-04-72f6f03af101) $ gcloud storage ls \
gs://coins_hun
gs://coins_hun/sample.txt
student_01_e80b229b3edb@cloudshell:~ (qwiklabs-gcp-04-72f6f03af101) $
```

Real World Application

- Give a developer read-only access to a specific storage bucket without project access
- Grant a contractor editor access temporarily then revoke when work is done
- Use service accounts with minimal IAM roles for automated processes
- Apply least privilege principle — only grant the permissions actually needed

What I Learned

- IAM roles control what users can and cannot do across GCP services
- Project-level roles like Viewer apply to all services in the project
- Resource-level roles like Storage Object Viewer apply to specific resources only
- Revoking a role can take up to 80 seconds to propagate
- Users with no project role can still access specific resources via resource-level roles
- The Grant Access button is grayed out for users without permission to manage IAM

Resources

GCP IAM Docs: <https://cloud.google.com/iam/docs>
Lab ID: GSP064